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Azure Enterprise / Principal Cloud Architect Playbook

Migration, Modernization & Greenfield — With AWS Comparisons

Author: Pushparaj Naik **Scope:** Microsoft Azure — End-to-End Lifecycle

Audience: Enterprise Architects, Principal Cloud Architects, Multi-Cloud Strategists

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Part 1 — Azure vs AWS: Philosophical Differences

1.1 Core Philosophy Comparison

Aspect	AWS	Azure
Design Philosophy	"Cloud-native first, broadest service catalog"	"Enterprise-first, hybrid by design, integrated with Microsoft ecosystem"
Identity Model	IAM (per-account, standalone)	Azure AD (Entra ID) — same identity for Office 365, Azure, and on-prem AD
Enterprise Appeal	Cloud-native companies, startups, mature cloud users	Enterprises already using Microsoft (AD, Office 365, SQL Server, .NET)
Hybrid Story	Outposts (limited), EKS Anywhere	Azure Arc (manage any infra from Azure), Azure Stack HCI/Hub
Developer Ecosystem	Polyglot (Python, Node, Go, Java equally supported)	.NET/C# first-class , but supports all (VS Code, GitHub integration)
AI/ML	Bedrock (managed LLMs), SageMaker	Azure OpenAI Service (exclusive GPT-4, GPT-4o access)
Licensing	Separate from enterprise software licensing	Azure Hybrid Benefit (reuse on-prem Windows/SQL licenses — 40-80% savings)

Aspect	AWS	Azure
Compliance	148 compliance certifications	100+ compliance offerings + deep government cloud (Azure Government, DoD)
Marketplace	Largest cloud marketplace	Strong enterprise ISV presence (SAP, Oracle, Citrix native integration)

1.2 When to Choose Azure Over AWS — The Architect's Decision

Scenario	Why Azure Wins	AWS Comparison
Microsoft shop (AD, O365, Teams, SharePoint)	Single identity (Entra ID), SSO across everything, Intune MDM	AWS has no equivalent Microsoft integration
Windows Server / SQL Server workloads	Azure Hybrid Benefit saves 40-80% by reusing on-prem licenses	AWS charges full license cost (BYOL is more complex)
.NET application portfolio	Azure App Service, Functions, and Visual Studio integration are best-in-class for .NET	AWS supports .NET but Azure is the native home
AI with GPT/OpenAI models	Azure OpenAI Service (exclusive enterprise access to GPT-4, GPT-4o, DALL-E)	AWS Bedrock has Claude, Titan, Llama — but no GPT-4
SAP workloads	SAP on Azure is a certified, co-engineered partnership	SAP on AWS works but Azure has deeper SAP partnership
Hybrid / on-premises integration	Azure Arc manages on-prem, multi-cloud, and edge from one pane	AWS Outposts is AWS-only, doesn't manage other clouds

Scenario	Why Azure Wins	AWS Comparison
Government / regulated industries	Azure Government (separate regions), DoD IL5/IL6, FedRAMP High	AWS GovCloud exists but Azure Government is more adopted in US federal
Enterprise Java modernization	Azure Spring Apps (managed Spring Boot) + Azure Kubernetes Service	AWS has no managed Spring Boot service

1.3 When AWS Still Wins Over Azure

Scenario	Why AWS Wins	Azure Gap
Cloud-native innovation	AWS innovates faster, more services, broader edge cases	Azure is more conservative, enterprise-focused
Kubernetes maturity	EKS is strong (but GKE is best)	AKS is good but GKE and EKS have more ecosystem tooling
Serverless maturity	Lambda + API Gateway + DynamoDB + Step Functions = best serverless stack	Azure Functions + Durable Functions are solid but less polished
IoT / Edge	IoT Core + Greengrass is the most complete edge stack	Azure IoT Hub + IoT Edge is close but Greengrass has more edge ML
Data Lake / Analytics	S3 + Glue + Athena + EMR is mature	Azure Data Lake Gen2 + Synapse is powerful but more complex
Global CDN reach	CloudFront with 450+ PoPs	Azure CDN / Front Door with ~120 PoPs

Part 2 — Azure Service Mapping (vs AWS)

2.1 Compute

Category	AWS	Azure	Key Difference
Virtual Machines	EC2	Azure Virtual Machines	Azure offers Hybrid Benefit (reuse Windows/SQL licenses — huge savings)
Managed Kubernetes	EKS	AKS (Azure Kubernetes Service)	AKS control plane is FREE (AWS EKS charges \$0.10/hr/cluster)
Serverless Containers	ECS Fargate / App Runner	Azure Container Apps	Container Apps = Dapr + KEDA integrated, event-driven by default
Serverless Functions	Lambda	Azure Functions	Azure Functions supports Durable Functions (stateful orchestrations)
PaaS (Web Apps)	Elastic Beanstalk	Azure App Service	App Service is far superior to Beanstalk — deployment slots, auto-scale, custom domains
Batch Processing	AWS Batch	Azure Batch	Similar capabilities
VM Scale Sets	Auto Scaling Groups	VM Scale Sets (VMSS)	Similar — VMSS supports Flexible orchestration mode
Spot/Low-Priority	Spot Instances	Spot VMs	Similar pricing model, Azure also has Low-Priority VMs for Batch
Dedicated Hosts	Dedicated Hosts	Azure Dedicated Host	Both offer single-tenant physical servers

2.2 Containers & Kubernetes

Category	AWS	Azure	Key Difference
Managed K8s	EKS	AKS	AKS control plane is FREE. EKS charges 0.10/hr(73/month)

Category	AWS	Azure	Key Difference
Container Registry	ECR	Azure Container Registry (ACR)	ACR supports geo-replication, content trust, Helm charts
Serverless Containers	App Runner / Fargate	Azure Container Apps	Container Apps has built-in Dapr, KEDA, traffic splitting
Container Instances	No direct equivalent	Azure Container Instances (ACI)	Run individual containers without clusters — great for burst/batch
Service Mesh	App Mesh / Istio on EKS	Istio on AKS (managed add-on)	AKS has managed Istio as first-class add-on
Multi-Cloud K8s	No equivalent	Azure Arc-enabled Kubernetes	Manage any K8s cluster from Azure (EKS, GKE, on-prem)
Spring Boot PaaS	No direct equivalent	Azure Spring Apps	Managed VMware Tanzu — Spring Boot apps with zero K8s ops

2.3 Database & Storage

Category	AWS	Azure	Key Difference
Object Storage	S3	Azure Blob Storage	Blob supports Hot/Cool/Cold/Archive tiers + immutable storage
Block Storage	EBS	Azure Managed Disks	Ultra Disk for high-IOPS, Premium SSD v2 for flexibility
File Storage	EFS	Azure Files	Azure Files supports SMB (Windows-native) + NFS
Managed SQL Server	RDS SQL Server	Azure SQL Database / SQL	Azure SQL is the native home for SQL Server (best compatibility + Hybrid Benefit)

Category	AWS	Azure	Key Difference
		Managed Instance	
Managed PostgreSQL	RDS / Aurora PostgreSQL	Azure Database for PostgreSQL Flexible Server	Flexible Server supports Citus (horizontal scale)
Managed MySQL	RDS / Aurora MySQL	Azure Database for MySQL Flexible Server	Similar, Azure has built-in HA
NoSQL (Document)	DynamoDB	Cosmos DB	Cosmos DB supports 5 APIs (SQL, MongoDB, Cassandra, Gremlin, Table)
Data Warehouse	Redshift	Azure Synapse Analytics	Synapse = data warehouse + data lake + Spark + pipelines in one
In-Memory Cache	ElastiCache	Azure Cache for Redis	Similar — Azure also offers Redis Enterprise tier
Distributed Database	Aurora Global Database	Cosmos DB (multi-region writes)	Cosmos DB supports 5 consistency levels (from Strong to Eventual)

2.4 Networking

Category	AWS	Azure	Key Difference
VPC	VPC (Regional)	VNet (Regional)	Both are regional. Azure supports VNet peering (global)
Load Balancer (L7)	ALB	Application Gateway	App Gateway includes WAF natively
Global Load Balancer	CloudFront + Route53	Azure Front Door	Front Door = global LB + CDN + WAF in one service

Category	AWS	Azure	Key Difference
Load Balancer (L4)	NLB	Azure Load Balancer	Similar
CDN	CloudFront	Azure CDN / Front Door	Front Door combines CDN + global LB + WAF
DNS	Route 53	Azure DNS	Azure DNS has Private DNS Zones (like Route53 PHZ)
VPN	Site-to-Site VPN	VPN Gateway	Similar — S2S, P2S, ExpressRoute
Dedicated Connectivity	Direct Connect	ExpressRoute	ExpressRoute supports Global Reach (connect on-prem sites through Microsoft backbone)
Hub-Spoke	Transit Gateway	Azure Virtual WAN / Hub-Spoke VNet	Virtual WAN = managed hub, or DIY hub-spoke with peering
Private Endpoints	VPC Endpoints (PrivateLink)	Azure Private Link / Private Endpoints	Same concept — private IPs for PaaS services
Firewall	AWS Network Firewall	Azure Firewall (Premium)	Azure Firewall Premium has TLS inspection, IDPS
DDoS	AWS Shield	Azure DDoS Protection	Standard tier auto-protects all public IPs in VNet
WAF	AWS WAF	Azure WAF (on App Gateway/Front Door)	WAF integrated into load balancing (simpler)

2.5 Security & Identity

Category	AWS	Azure	Key Difference
Identity Provider	IAM (cloud only)	Microsoft Entra ID (Azure AD)	Entra ID = enterprise IdP for Azure, O365, SaaS apps, on-prem AD sync
SSO	IAM Identity Center	Entra ID SSO	Entra ID provides SSO for 3000+ SaaS apps (Salesforce, ServiceNow, etc.)
MFA / Conditional Access	IAM MFA	Entra ID Conditional Access	Risk-based, device-aware, location-aware policies (far richer than AWS MFA)
Privileged Identity	No equivalent	Entra PIM (Privileged Identity Management)	Just-in-time admin access with approval workflows
Secrets	Secrets Manager	Azure Key Vault	Key Vault combines secrets + keys + certificates (AWS splits these into 3 services)
Threat Detection	GuardDuty	Microsoft Defender for Cloud	Defender covers Azure, AWS, GCP, on-prem (multi-cloud security)
Security Posture	Security Hub	Microsoft Defender for Cloud (CSPM)	Includes Secure Score, compliance dashboard, recommendations
SIEM	No native SIEM (use third-party)	Microsoft Sentinel	Cloud-native SIEM + SOAR — collects from Azure, O365, AWS, GCP
Organization Policies	SCPs	Azure Policy	Azure Policy = preventive + detective. Deny, audit, modify, deploy-if-not-exists
Governance Framework	Control Tower	Azure Landing Zones (ALZ) + Management Groups	Management Groups = hierarchical policy inheritance

2.6 DevOps & CI/CD

Category	AWS	Azure	Key Difference
DevOps Platform	CodePipeline + CodeBuild (or GitHub Actions)	Azure DevOps (Boards, Repos, Pipelines, Artifacts, Test Plans)	Azure DevOps is a complete ALM platform (not just CI/CD)
CI/CD Pipeline	CodePipeline + CodeBuild	Azure Pipelines (or GitHub Actions)	Azure Pipelines has YAML + Classic (visual) editor, multi-stage
Artifact Storage	ECR + CodeArtifact	Azure Artifacts + ACR	Azure Artifacts supports npm, NuGet, Maven, Python, Universal
Source Control	CodeCommit (deprecated)	Azure Repos (or GitHub)	Microsoft owns GitHub — deep integration
Project Management	No equivalent	Azure Boards	Boards = Agile project management (backlogs, sprints, Kanban)
Test Management	No equivalent	Azure Test Plans	Integrated test case management
Monitoring	CloudWatch	Azure Monitor	Azure Monitor = metrics + logs + alerts + Application Insights (APM)
APM	X-Ray	Application Insights	App Insights has live metrics, smart detection, availability tests

Category	AWS	Azure	Key Difference
Log Analytics	CloudWatch Logs	Log Analytics Workspace	KQL (Kusto Query Language) is more powerful than CloudWatch Insights
IaC	CloudFormation / Terraform	Bicep / ARM Templates / Terraform	Bicep = Azure-native IaC (transpiles to ARM), simpler than ARM JSON

2.7 AI/ML & Data

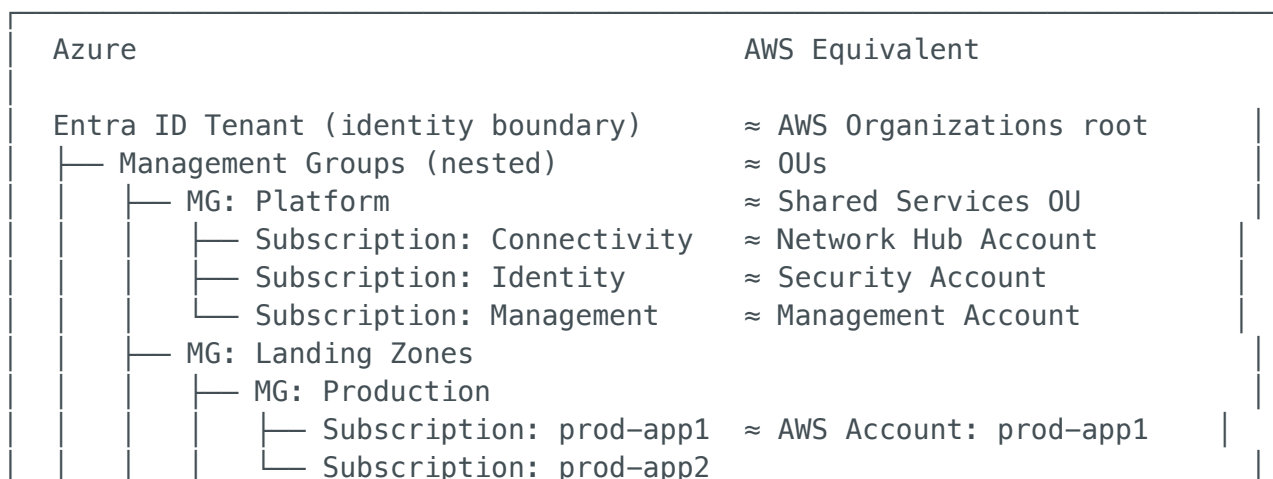
Category	AWS	Azure	Key Difference
LLM Platform	Bedrock (Claude, Titan, Llama)	Azure OpenAI Service	Exclusive enterprise access to GPT-4, GPT-4o, o1, DALL-E, Whisper
ML Platform	SageMaker	Azure Machine Learning	AML Studio has designer (low-code) + notebooks + pipelines
Cognitive Services	Rekognition, Comprehend, Polly (separate)	Azure AI Services (unified SDK)	Vision, Speech, Language, Decision — single SDK, single endpoint
RAG / Search	Bedrock Knowledge Bases + OpenSearch	Azure AI Search (Cognitive Search)	AI Search = vector + keyword + semantic search in one
Data Warehouse	Redshift	Synapse Analytics (Dedicated SQL Pool)	Synapse = warehouse + lake + Spark + pipelines unified
Data Lake	S3 + Glue + Lake Formation	Azure Data Lake Storage Gen2	ADLS Gen2 = hierarchical namespace

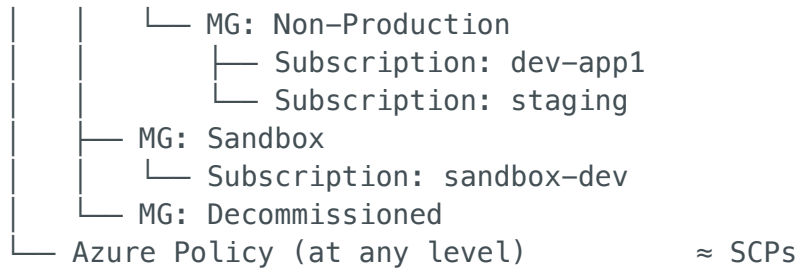
Category	AWS	Azure	Key Difference
			on Blob Storage (HDFS-compatible)
Stream Processing	Kinesis	Event Hubs + Stream Analytics	Event Hubs ≈ Kafka (has Kafka-compatible endpoint!)
ETL / Data Integration	Glue	Azure Data Factory	Data Factory = visual ETL, 100+ connectors, mapping data flows
Messaging	SQS + SNS	Service Bus + Event Grid	Service Bus = enterprise messaging (sessions, dead-letter, transactions)
Workflow	Step Functions	Logic Apps / Durable Functions	Logic Apps = visual workflow designer with 400+ connectors

Part 3 — Azure Organization & Landing Zone

3.1 Azure Resource Hierarchy — Fundamentally Different from AWS

AZURE RESOURCE HIERARCHY (vs AWS):

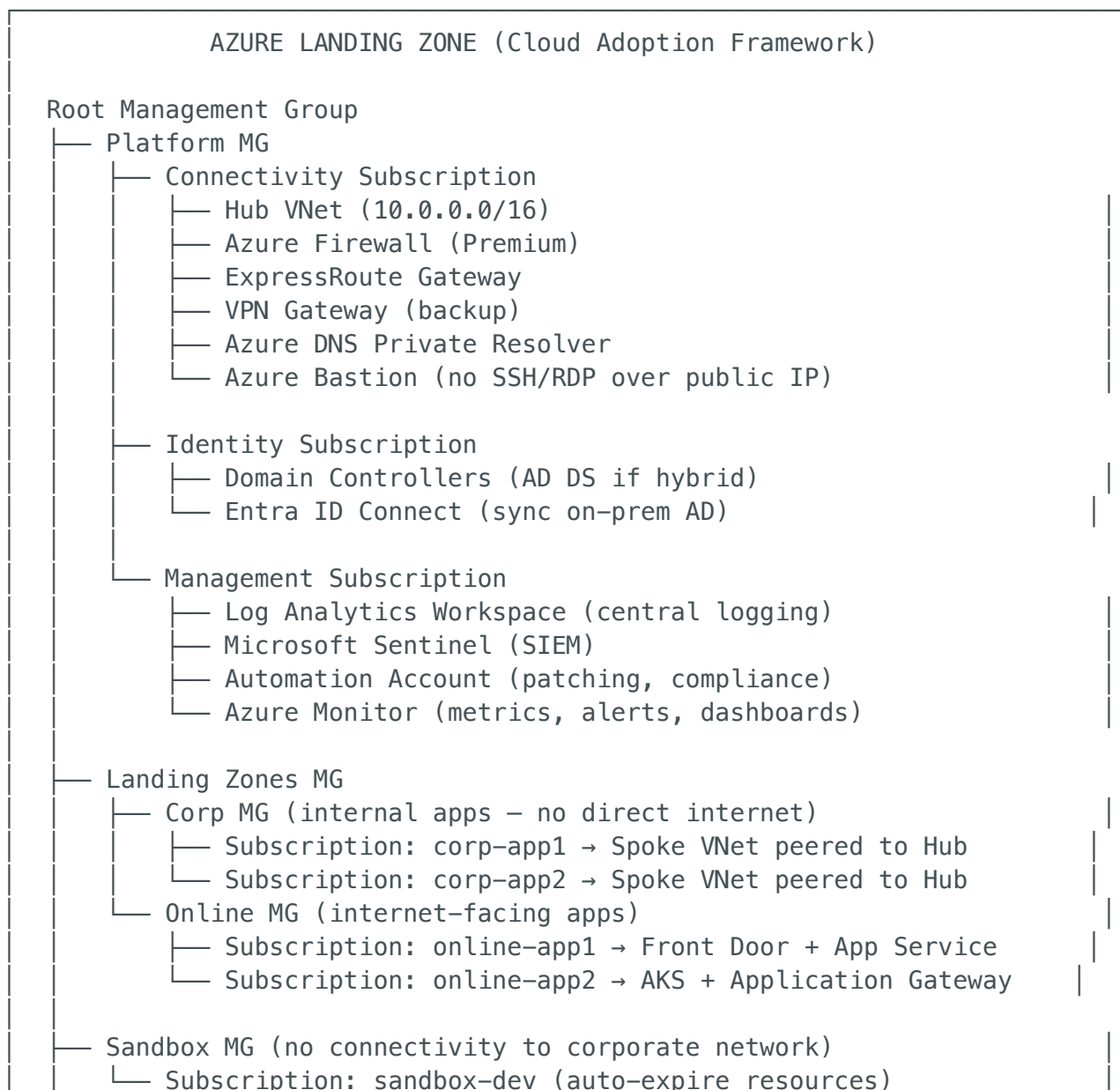




KEY DIFFERENCES FROM AWS:

- Azure "Subscription" = AWS "Account" (blast radius boundary)
- Azure "Management Group" = AWS "OU" (but unlimited nesting)
- Azure identity is Entra ID (shared with 0365, on-prem AD)
- Azure Policy is MORE POWERFUL than SCPs (can audit, modify, deploy)
- Azure has Resource Groups WITHIN subscriptions (no AWS equivalent)
- Azure RBAC inherits down: MG → Subscription → RG → Resource

3.2 Azure Landing Zone Architecture (CAF)



Decommissioned MG (migrated/retired subscriptions)

POLICY ASSIGNMENTS (at Root MG level):

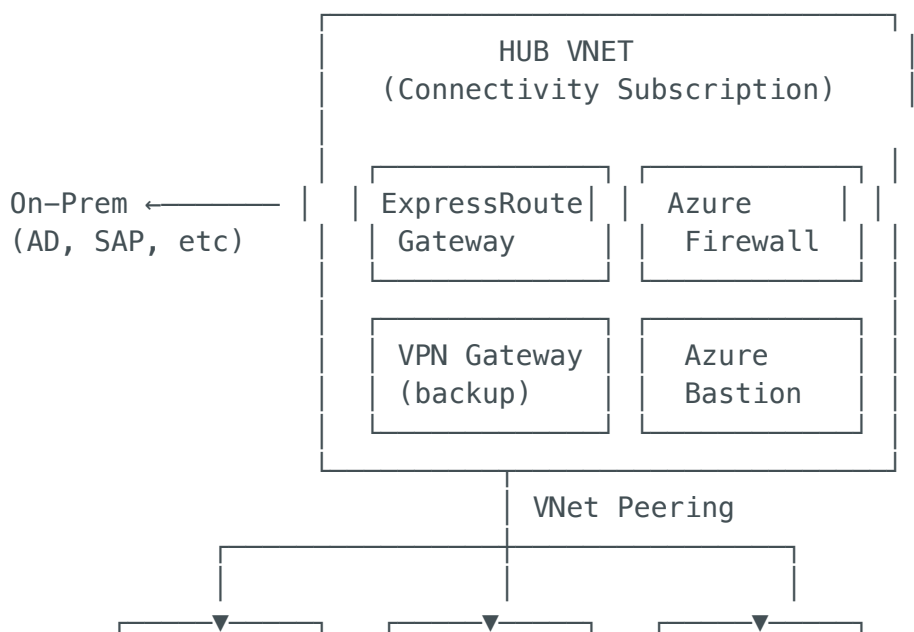
- Require resource tagging (CostCenter, Owner, Environment)
- Deny public IP creation on VMs
- Require encryption on all storage accounts
- Audit/Deny resources outside approved regions
- Deploy Defender for Cloud on all subscriptions
- Deploy diagnostic settings to Log Analytics
- Deny classic/legacy resources

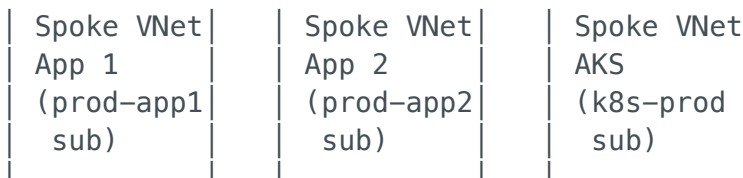
AWS COMPARISON:

Root MG	≈ Organizations root
Platform MG	≈ Shared Services OU (Security, Network, Logging accounts)
Connectivity Sub	≈ Network Hub Account (Transit Gateway, Direct Connect)
Management Sub	≈ Log Archive Account + Management Account
Landing Zones MG	≈ Workload OUs (Prod, Non-Prod)
Azure Policy	≈ SCPs (but Azure Policy is much more powerful)
Azure Firewall	≈ AWS Network Firewall (Azure Firewall is more mature)
ExpressRoute	≈ Direct Connect
Azure Bastion	≈ SSM Session Manager (no bastion host needed)
Log Analytics Workspace	≈ CloudWatch Logs (but KQL >> CloudWatch Insights)
Microsoft Sentinel	≈ No AWS equivalent (GuardDuty + Security Hub + 3rd party SIEM)

3.3 Hub-Spoke Network Topology

AZURE HUB-SPOKE (vs AWS Transit Gateway):





AWS EQUIVALENT:

Hub VNet + Azure Firewall	≈	Inspection VPC + AWS Network Firewall + Transit Gateway
VNet Peering (simpler)	≈	TGW attachments (but VNet peering is simpler)
ExpressRoute	≈	Direct Connect
Azure Bastion	≈	SSM Session Manager

KEY AZURE ADVANTAGE:

- Azure Firewall is more mature (TLS inspection, IDPS, threat intel, web categories)
- Azure Bastion = managed jump box (no EC2 bastion host needed)
- VNet peering is simpler than Transit Gateway for <50 spokes
- For >50 spokes, use Azure Virtual WAN (≈ managed Transit Gateway)

3.4 Landing Zone Automation

Tool	Azure	AWS Equivalent
Landing Zone Blueprint	Azure Landing Zone Accelerator (Bicep/Terraform)	Control Tower
Policy Framework	Azure Policy (built-in initiative definitions)	SCPs + Config Rules
Subscription Vending	Subscription Vending Module (Terraform)	Control Tower Account Factory
IaC (Azure-native)	Bicep (cleaner than ARM JSON)	CloudFormation
IaC (multi-cloud)	Terraform (AzureRM provider)	Terraform (AWS provider)
Governance Dashboard	Azure Resource Graph + Workbooks	Config + Security Hub

Part 4 — Azure Migration: Assess → Migrate → Modernize → Optimize

4.1 Azure Migration Framework (Cloud Adoption Framework)

AZURE CLOUD ADOPTION FRAMEWORK (CAF) PHASES:

AWS MAP: Assess → Mobilize → Migrate & Modernize

Azure CAF: Strategy → Plan → Ready → Migrate/Innovate → Govern → Manage

Azure CAF is BROADER than AWS MAP – it includes governance and management as ongoing phases, not just migration checkpoints.

Strategy: Define business justifications and expected outcomes

Plan: Align actionable adoption plan with business outcomes

Ready: Prepare the cloud environment (Landing Zone)

Migrate: Migrate existing workloads

Innovate: Build new cloud-native solutions (Greenfield)

Govern: Govern the environment (policies, cost, security)

Manage: Manage operations (monitoring, patching, DR)

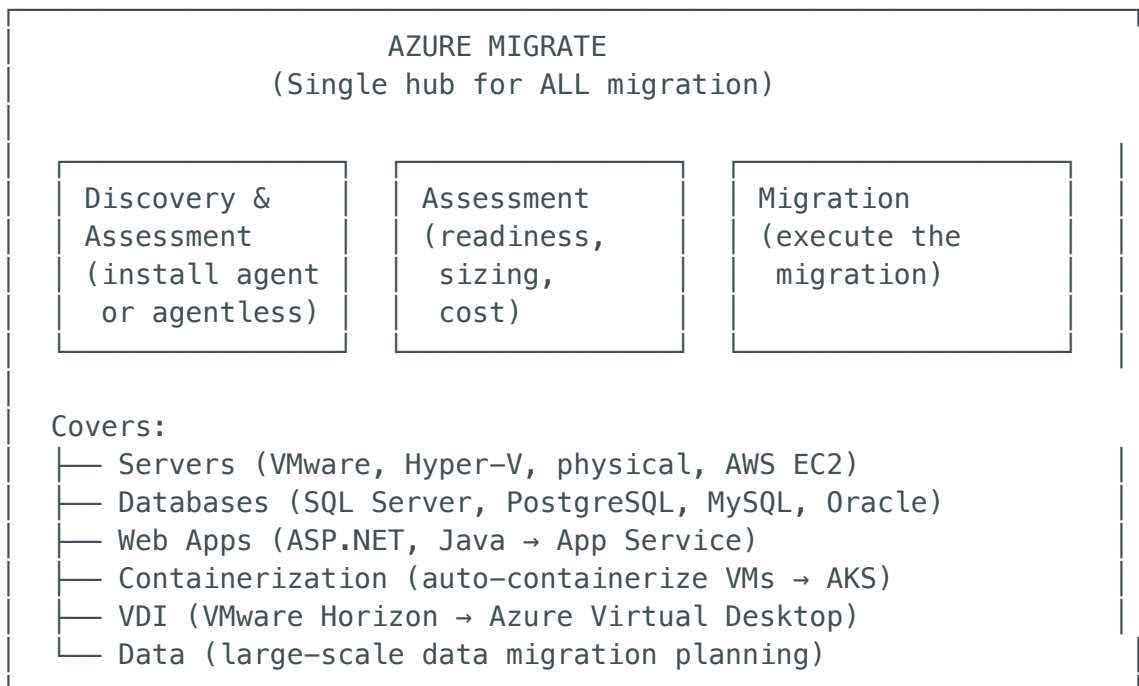
4.2 Azure Migration Tools (vs AWS)

Migration Task	Azure Tool	AWS Equivalent
Portfolio Discovery	Azure Migrate (unified hub)	Application Discovery Service + Migration Hub
VM Migration	Azure Migrate: Server Migration	Application Migration Service (MGN)
Database Migration	Azure Database Migration Service (DMS)	AWS DMS
Schema Assessment	Data Migration Assistant (DMA)	Schema Conversion Tool (SCT)
Web App Migration	Azure Migrate: App Service Migration	No equivalent (manual)
Containerization	Azure Migrate: App Containerization	No equivalent (manual)
Large Data Transfer	Azure Data Box (100TB/1PB)	Snowball / Snowmobile

Migration Task	Azure Tool	AWS Equivalent
Online Data Transfer	AzCopy / Azure File Sync	DataSync
SQL Server Assessment	Azure SQL Migration Extension	No equivalent
Cost Estimation	TCO Calculator + Azure Migrate	Migration Evaluator

4.3 Azure Migrate – The Unified Migration Hub

AZURE MIGRATE (vs AWS – which splits across multiple tools):



AWS EQUIVALENT REQUIRES MULTIPLE TOOLS:

Azure Migrate (discovery)	= Application Discovery Service
Azure Migrate (assessment)	= Migration Evaluator (TSO Logic)
Azure Migrate (VM migration)	= Application Migration Service (MGN)
Azure Migrate (DB migration)	= DMS + SCT
Azure Migrate (web apps)	= No equivalent (manual)
Azure Migrate (containers)	= No equivalent (manual)

AZURE ADVANTAGE: ONE tool to discover, assess, and migrate everything.
AWS requires 4–5 separate tools for the same scope.

4.4 Database Migration — Azure Specifics

DATABASE MIGRATION ON AZURE:

Source Database	→ Azure Target	→ AWS Equivalent
SQL Server (closest: RDS)	→ Azure SQL Database (PaaS) → SQL Managed Instance (PaaS, 100%) → SQL Server on Azure VM (IaaS)	→ RDS SQL Server → No equivalent → EC2 + SQL
Oracle	→ Azure SQL (with conversion)	→ Aurora
PostgreSQL	→ Azure DB for PostgreSQL	→ Aurora
PostgreSQL	→ Oracle on Azure VM	→ Oracle on EC2
PostgreSQL PostgreSQL	→ Azure DB for PostgreSQL Flex	→ RDS/Aurora
MySQL MySQL	→ Azure DB for MySQL Flexible	→ RDS/Aurora
MongoDB on AWS	→ Cosmos DB (MongoDB API) → MongoDB Atlas on Azure	→ DocumentDB → MongoDB Atlas
Cassandra (managed Cassandra)	→ Cosmos DB (Cassandra API)	→ Keyspaces

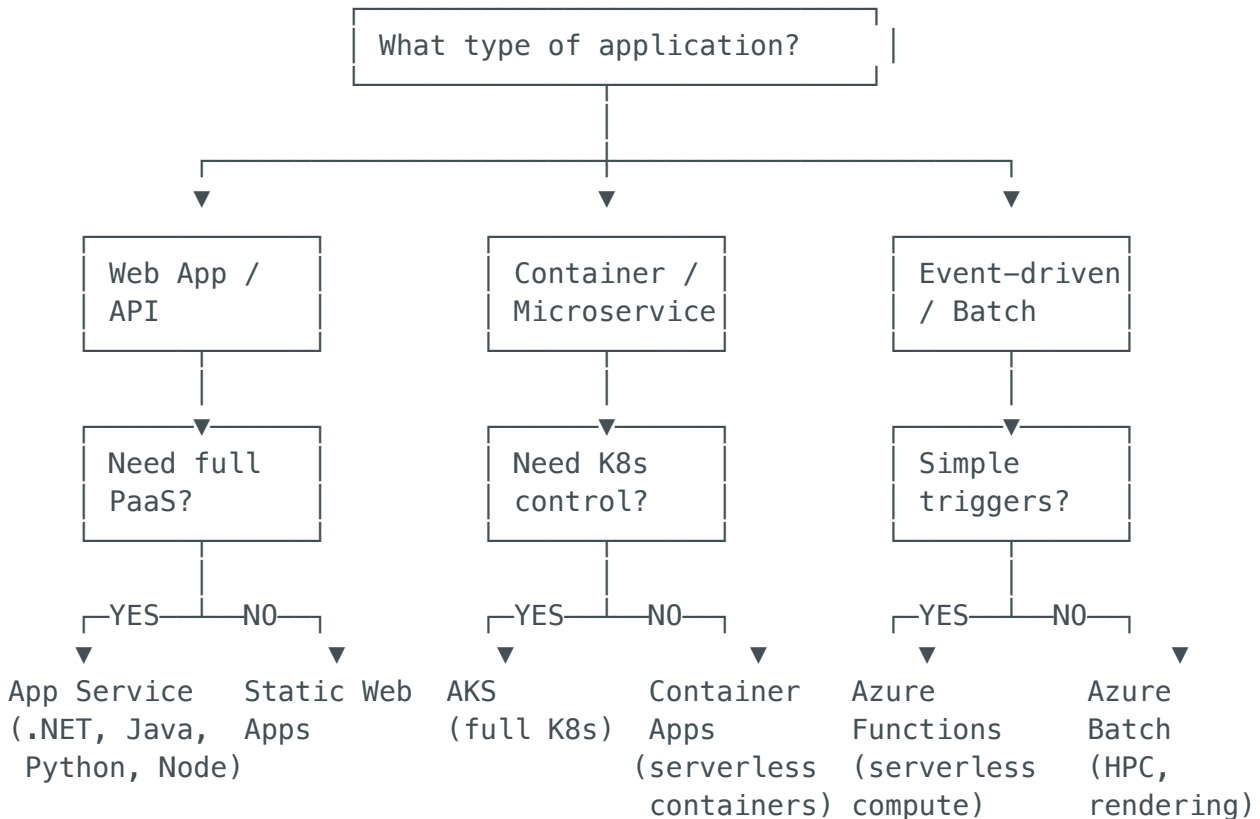
AZURE UNIQUE ADVANTAGES:

1. SQL Managed Instance = 100% SQL Server compatibility in PaaS
(No AWS equivalent – RDS SQL Server has limitations)
2. Azure Hybrid Benefit = Reuse on-prem SQL Server licenses
(Saves 40–55% vs full-price Azure SQL, no AWS equivalent)
3. Azure SQL has built-in:
 - └ Intelligent Performance (auto-tuning indexes, query store)
 - └ Hyperscale tier (100TB+, instant scale, rapid backup)
 - └ Serverless tier (auto-pause, auto-scale, pay per second)
 - └ Elastic Pools (share resources across databases)
4. Cosmos DB supports 5 APIs in one service:
SQL (document), MongoDB, Cassandra, Gremlin (graph), Table
(AWS has separate services: DynamoDB, DocumentDB, Keyspaces, Neptune)

Part 5 — Azure Modernization & Cloud-Native Patterns

5.1 Compute Decision Tree on Azure

APPLICATION MODERNIZATION — WHERE TO RUN IT ON AZURE:



AWS COMPARISON:

App Service	≈ Elastic Beanstalk (but App Service is FAR superior)
Static Web Apps	≈ Amplify Hosting
AKS	≈ EKS (AKS control plane is FREE)
Container Apps	≈ App Runner + Fargate (Container Apps has Dapr + KEDA built-in)
Azure Functions	≈ Lambda (Functions supports Durable Functions for stateful)
Azure Batch	≈ AWS Batch

5.2 AKS Deep Dive — vs EKS

Feature	AKS (Azure)	EKS (AWS)	Winner
Control Plane Cost	FREE	0.10/hr(73/month per cluster)	AKS
Managed Upgrades	Auto-upgrade channels (Rapid, Stable)	Manual upgrade process	AKS
Managed Istio	Istio-based service mesh (add-on)	App Mesh or manual Istio	Tie
Workload Identity	Azure AD Workload Identity	IRSA	Similar
Windows Containers	Full Windows node pool support	Limited Windows support	AKS
Virtual Nodes (serverless)	ACI integration (burst to serverless)	Fargate profiles	Similar
GitOps	Flux (GitOps add-on)	ArgoCD (third-party)	Tie
Dev Spaces	Bridge to Kubernetes (VS Code)	No equivalent	AKS
Multi-Cluster	Azure Arc-enabled K8s	No equivalent	AKS
Node Autoscaling	Cluster Autoscaler + KEDA	Karpenter / Cluster Autoscaler	EKS (Karpenter)

5.3 Azure App Service – The .NET/Java PaaS Champion

APP SERVICE (No real AWS equivalent – Beanstalk is far inferior):

Features that AWS Elastic Beanstalk DOESN'T have:

- └─ Deployment Slots (blue/green with traffic routing by percentage)
- └─ Auto-heal (restart app on memory leak, slow responses)
- └─ VNet Integration (private access to databases without NAT)
- └─ Hybrid Connections (connect to on-prem databases without VPN)
- └─ Managed Certificates (auto-SSL with custom domains)
- └─ Authentication built-in (Entra ID, Google, Facebook – zero code)
- └─ WebJobs (background processing attached to the app)
- └─ Azure Spring Apps (managed VMware Tanzu for Spring Boot)
- └─ Automatic OS patching (no maintenance window)

WHEN TO USE APP SERVICE:

- .NET (ASP.NET Core, .NET Framework) web apps
- Java (Spring Boot, Tomcat, JBoss) web apps
- Node.js / Python / PHP / Ruby web apps
- REST APIs
- Mobile backends
- Any HTTP workload that doesn't need containers/K8s complexity

5.4 Cosmos DB — Azure's Unique Global Database

COSMOS DB vs AWS DATABASES:

COSMOS DB CAPABILITIES

5 APIs in ONE Service:

- NoSQL (native – SQL-like query over JSON documents)
- MongoDB API (drop-in replacement for MongoDB)
- Cassandra API (drop-in replacement for Cassandra)
- Gremlin API (graph database)
- Table API (key-value, replaces Azure Table Storage)

Unique Features:

- 5 Consistency Levels:
Strong → Bounded Staleness → Session → Consistent Prefix → Eventual
(AWS DynamoDB only offers: Strong or Eventual – no middle ground)
- Multi-region writes (active-active globally)
- Automatic indexing (every property indexed, no management)
- Serverless tier (pay per operation) OR Provisioned (RU/s)
- < 10ms reads, < 10ms writes (SLA-backed)
- 99.999% availability SLA (with multi-region writes)

AWS EQUIVALENTS (requires 4 separate services):

- DynamoDB (NoSQL / key-value)
- DocumentDB (MongoDB compatible)
- Keyspaces (Cassandra compatible)
- Neptune (Graph – Gremlin/SPARQL)

COSMOS DB ADVANTAGE: One service, one SDK, one monitoring, one billing model – vs 4 separate AWS services with different APIs, SDKs, pricing models, and operational models.

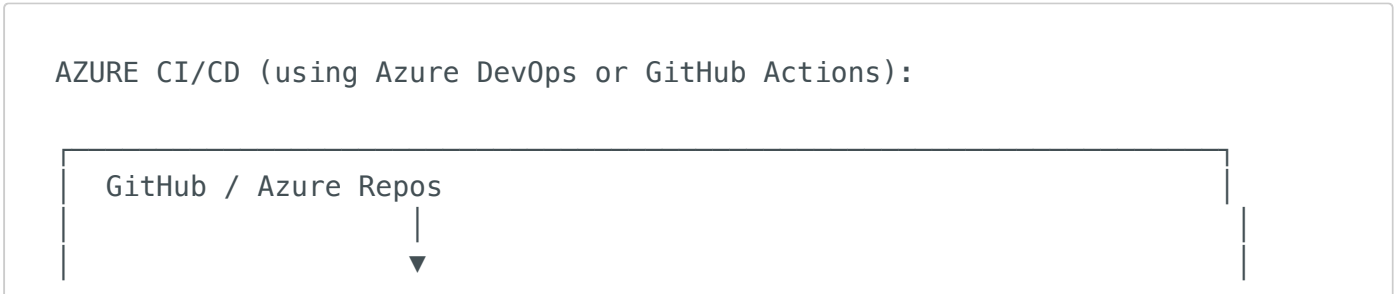
Part 6 — Azure DevOps & CI/CD

Architecture

6.1 Azure DevOps vs GitHub Actions vs AWS

Capability	Azure DevOps Pipelines	GitHub Actions	AWS CodePipeline/Build
Visual Editor	Yes (Classic)	No	Yes (console only)
YAML Pipeline	Yes (multi-stage)	Yes	Limited
Self-Hosted Agents	Yes	Yes (runners)	No
Approval Gates	Yes (environments)	Yes (environments)	Yes (manual approval)
Artifact Management	Azure Artifacts (NuGet, npm, Maven, Python)	GitHub Packages	CodeArtifact
Board/Work Tracking	Azure Boards (Agile, Scrum, CMMI)	GitHub Issues/Projects	No equivalent
Test Management	Azure Test Plans	No built-in	No equivalent
Integration	Azure + GitHub + any Git	GitHub + any cloud	AWS only
Cost	Free for first 5 users, 1800 min CI/CD	Free 2000 min/month (public repos)	Pay per build minute

6.2 Azure CI/CD Pipeline Architecture



Azure Pipelines / GitHub Actions

Stage 1: BUILD

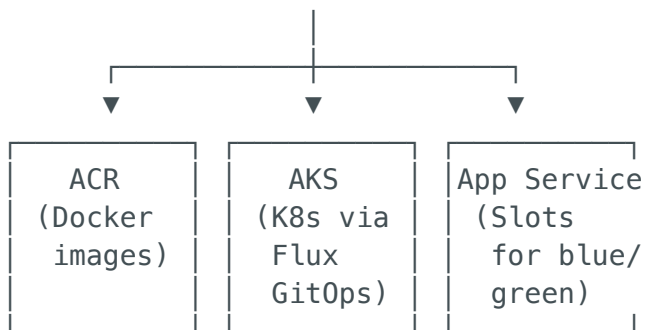
- Restore dependencies
- Compile / build
- Unit tests
- SonarQube / SAST
- Docker build

Stage 2: SCAN

- Trivy (container scan)
- OWASP DC (dependency scan)
- Checkov / tfsec (IaC scan)
- Push to ACR

Stage 3: DEPLOY (per environment)

- DEV: auto-deploy
- STAGING: auto-deploy + tests
- PROD: manual approval → deploy



6.3 Azure Monitoring & Observability

Capability	Azure Service	AWS Equivalent	Key Difference
Metrics	Azure Monitor Metrics	CloudWatch Metrics	Similar
Logs	Log Analytics (KQL)	CloudWatch Logs (Insights)	KQL is far more powerful than CloudWatch Insights
APM	Application Insights	X-Ray	App Insights is a complete APM with live metrics, auto-detection
Distributed Tracing	Application Insights	X-Ray	App Insights has end-to-end transaction tracking

Capability	Azure Service	AWS Equivalent	Key Difference
Smart Detection	App Insights Anomaly Detection	No equivalent	Automatically detects failures, latency anomalies
Live Metrics	App Insights Live Metrics	No equivalent	Real-time streaming of telemetry
Availability Tests	App Insights URL Ping/Multi-step	Route 53 Health Checks	More sophisticated web test scenarios
Dashboards	Azure Dashboards + Workbooks	CloudWatch Dashboards	Workbooks have interactive, parametric queries
SIEM	Microsoft Sentinel	No native SIEM	Sentinel = cloud SIEM + SOAR + threat hunting
Alerting	Azure Monitor Alerts	CloudWatch Alarms	Action Groups route to email, SMS, Logic App, Function, webhook

Part 7 — Azure Security, Compliance & Governance

7.1 Security Architecture — Azure vs AWS

AZURE SECURITY MODEL vs AWS:

Azure

AWS

Azure Policy (preventive + detective)

├ Deny (≈ SCP deny)

├ Audit (≈ Config Rule audit)

├ Modify (auto-tag, auto-remediate)

├ DeployIfNotExists (auto-deploy CFn)

└ Append (add fields to requests)

Microsoft Defender for Cloud

SCPs (preventive only)

├ Deny

├ (SCPs can only deny, not

├ (No equivalent to Modify)

├ (No equivalent – must use

└

GuardDuty + Security Hub +

Inspector

— Secure Score (0-100 score)	— Security Hub score
— CSPM (multi-cloud – AWS, GCP too!)	— AWS-only CSPM
— Defender for Servers (EDR)	— GuardDuty (no EDR)
— Defender for Containers	— No container security
— Defender for SQL	— No database security
— Defender for Storage	— Macie (S3 only)
— Defender for Key Vault	— No KMS monitoring
— Regulatory compliance dashboard	— Config Conformance Packs

Microsoft Sentinel (SIEM + SOAR)

— 200+ data connectors	— No native SIEM
— KQL-based detection rules	— (Need Splunk/Datadog/etc.)
— Automated playbooks (Logic Apps)	
— Threat hunting notebooks	
— UEBA (User Entity Behavior Analytics)	
— Collects from Azure, 0365, AWS, GCP	

AZURE UNIQUE SECURITY ADVANTAGES:

1. Azure Policy is MORE POWERFUL than SCPs
(SCPs can only deny. Azure Policy can deny, audit, modify, deploy-if-not-exists)
2. Microsoft Defender for Cloud works MULTI-CLOUD
(Protects AWS and GCP workloads too – not just Azure)
3. Microsoft Sentinel is a cloud-native SIEM
(AWS has no native SIEM – must use third-party)
4. Entra ID Conditional Access
(Risk-based, device-aware, location-aware access policies – far beyond AWS MFA)
5. Entra PIM (Privileged Identity Management)
(Just-in-time admin access with approval workflows – no AWS equivalent)

7.2 Azure Policy – More Powerful Than AWS SCPs

AZURE POLICY EFFECTS (vs AWS SCPs):

Effect	Description	AWS Equivalent
Deny	Block non-compliant resource creation	SCP Deny
Audit	Log non-compliance (don't block)	Config Rule (separate svc)
Modify	Auto-fix resources	NO EQUIVALENT

	(add tags, fix settings)	
DeployIfNotExists	Auto-deploy missing resources (e.g., deploy diagnostic settings)	NO EQUIVALENT
Append	Add fields to requests (e.g., force HTTPS)	NO EQUIVALENT
AuditIfNotExists	Audit if related resource is missing	NO EQUIVALENT
Disabled	Turn off a policy	N/A

EXAMPLE – Auto-deploy diagnostic settings:

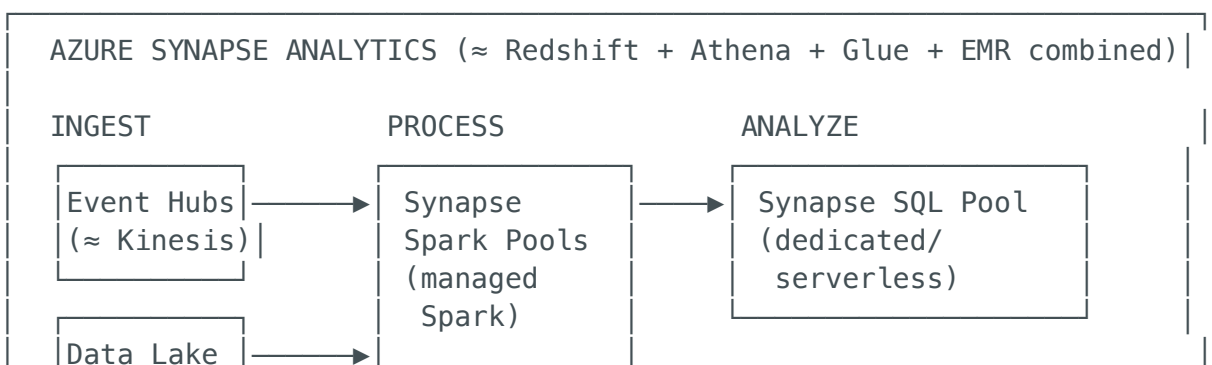
```
{
  "effect": "DeployIfNotExists",
  "details": {
    "type": "Microsoft.Insights/diagnosticSettings",
    "deployment": {
      // ARM template to create diagnostic settings
      // sending logs to central Log Analytics Workspace
    }
  }
}
```

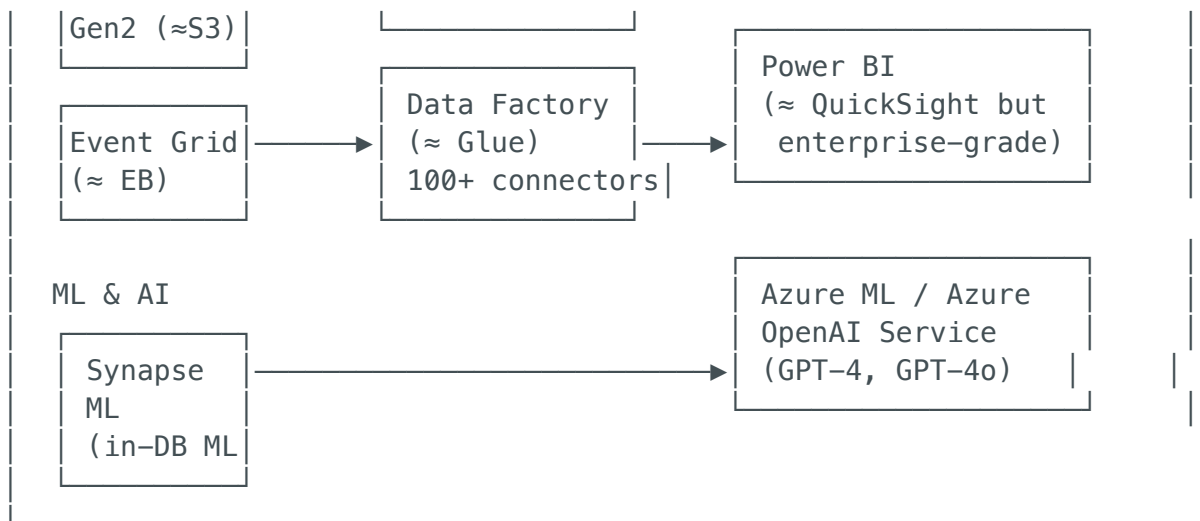
This AUTOMATICALLY configures logging for EVERY new resource.
On AWS, you'd need a CloudTrail + Lambda + Config Rule chain to do this.

Part 8 – Azure Data & AI Platform Architecture

8.1 Azure Data Platform Architecture

AZURE DATA PLATFORM (vs AWS):





AWS EQUIVALENT REQUIRES SEPARATE SERVICES:

Synapse SQL Pool	= Redshift (dedicated) + Athena (serverless)
Synapse Spark	= EMR (Spark)
Data Factory	= Glue
Event Hubs	= Kinesis Data Streams
Data Lake Gen2	= S3 + Lake Formation
Power BI	= QuickSight (Power BI is more powerful/adopted)
Synapse Pipelines	= Glue Workflows / Step Functions

Azure: 1 service (Synapse) with integrated workspace

AWS: 5+ separate services, separate consoles, separate billing

8.2 Azure OpenAI Service — Azure's AI Differentiator

AZURE OPENAI SERVICE vs AWS BEDROCK:

Azure OpenAI Service	AWS Bedrock
GPT-4, GPT-4o, GPT-4-turbo	Claude 3.5 (Anthropic)
o1, o1-mini (reasoning)	Titan (Amazon)
DALL-E 3 (images)	Llama 3 (Meta)
Whisper (speech-to-text)	Mistral, Cohere
Text Embedding (ada-002)	Titan Embeddings
GPT-4 Vision (multimodal)	Claude 3 Vision
AZURE ADVANTAGE:	
— GPT-4 is the market leader – Azure has exclusive access	
— Enterprise features: content filtering, abuse monitoring	
— Runs in YOUR Azure subscription (data stays in region)	
— Private endpoints – no data leaves your VNet	
— Azure AI Search integration for RAG	
— Same Azure RBAC, Azure Policy, compliance framework	
AWS ADVANTAGE:	

- └─ More model VARIETY (Claude, Llama, Mistral, Cohere)
- └─ Fine-tuning for more models
- └─ Knowledge Bases (managed RAG)
- └─ Agents (multi-step reasoning)

ARCHITECT'S TAKE:

- If you need GPT-4/o1 specifically → Azure OpenAI
- If you want model diversity/flexibility → AWS Bedrock
- If you want latest Google models → GCP Vertex AI (Gemini)

Part 9 — Azure Networking Deep Dive

9.1 VNet Architecture — Regional Like AWS

AZURE VNET vs AWS VPC:

Both are REGIONAL (unlike GCP's Global VPC):

Azure VNet:

- └─ Regional (same as AWS)
- └─ Subnets are AZ-spanning by default (no need to create subnet-per-AZ)
- └─ NSGs attach to NIC or Subnet
- └─ ASGs (Application Security Groups)
- └─ Service Endpoints (route to PaaS gateway/interface)
- └─ Private Endpoints (Private Link)
- └─ VNet Peering (global, non-transitive)
- └─ Azure Firewall (NVA alternative)

AWS VPC:

- └─ Regional
- └─ Subnets are AZ-specific (must create subnet per AZ)
- └─ SGs attach to ENI
- └─ No equivalent (tag-based SGs)
- └─ VPC Endpoints
- └─ PrivateLink
- └─ VPC Peering (global, non-transitive)
- └─ AWS Network Firewall

KEY DIFFERENCE — SUBNETS:

Azure: Subnets span ALL availability zones in a region.
One subnet = all AZs. Simpler! Fewer subnets to manage.

AWS: Subnets are AZ-specific.
Need 3 subnets × 3 tiers = 9 subnets minimum.
More granular but more complex.

9.2 Azure Front Door — Global Load Balancing + CDN + WAF

AZURE FRONT DOOR (vs AWS CloudFront + ALB + WAF + Route53):

Azure Front Door = ONE SERVICE that combines:

— Global load balancing (Layer 7)	= CloudFront + Route53
— CDN (content caching at edge)	= CloudFront
— WAF (web application firewall)	= AWS WAF
— SSL offloading	= ACM + ALB
— URL-based routing	= ALB listener rules
— Session affinity	= ALB sticky sessions
— Health probes with auto-failover	= Route53 health checks
— Caching rules	= CloudFront behaviors
— Private Link to origins	= PrivateLink + ALB

AWS requires 4+ separate services to achieve what Front Door does in one.

WHEN TO USE FRONT DOOR:

- Internet-facing applications needing global reach
- Multi-region deployments with automatic failover
- Applications needing WAF + CDN + LB in one
- API backends behind a global endpoint

9.3 ExpressRoute vs AWS Direct Connect

Feature	Azure ExpressRoute	AWS Direct Connect
Speed	50 Mbps – 100 Gbps	1 Gbps – 100 Gbps
Global Reach	Connect on-prem sites through Microsoft backbone	No equivalent (need separate circuits)
Microsoft Peering	Access O365, Dynamics via private circuit	No equivalent (O365 is Microsoft)
Redundancy	Zone-redundant gateways, ExpressRoute circuits	Redundant connections to DX locations
Encryption	MACsec (layer 2 encryption)	MACsec
Pricing	Metered (\$0.025/GB outbound) or Unlimited	Port hour + data transfer

Part 10 — Azure Cost Management & FinOps

10.1 Azure Pricing Model (vs AWS)

Pricing Feature	Azure	AWS
On-Demand	Pay-as-you-go (per-second for Linux, per-minute for Windows)	Per-second billing
Reserved Instances	1-year (≈40% off) or 3-year (≈60% off)	1-year or 3-year RIs
Savings Plans	Azure Savings Plans (compute, similar to AWS)	AWS Savings Plans
Spot	Spot VMs (up to 90% off)	Spot Instances (similar)
Hybrid Benefit	Reuse Windows/SQL Server licenses — 40-80% savings	No equivalent (BYOL is limited)
Dev/Test Pricing	Azure Dev/Test subscriptions — discounted rates, no license fees	No equivalent
Free Tier	12-month free + always-free services	12-month free + always-free
Enterprise Agreement	EA pricing, MACC (Microsoft Azure Consumption Commitment)	EDP (Enterprise Discount Program)

10.2 Azure Cost Management Tools

Tool	Azure	AWS Equivalent
Cost Analysis	Cost Management + Billing (native, free)	Cost Explorer
Budgets	Azure Budgets (with action groups)	AWS Budgets
Advisor Recommendations	Azure Advisor (cost, security, reliability, performance, operational excellence)	Trusted Advisor
Right-Sizing	Azure Advisor VM right-sizing	Compute Optimizer

Tool	Azure	AWS Equivalent
Reservation Recommendations	Azure Advisor RI recommendations	RI/SP recommendations
Cost Allocation	Tags + Cost Management scopes	Tags + Cost Allocation Tags
Exports	Cost Management exports to Storage Account	CUR export to S3
Third-Party	Finout, CloudHealth, Spot.io	Same third-party tools

10.3 Azure Hybrid Benefit — The Enterprise Cost Saver

AZURE HYBRID BENEFIT (NO AWS EQUIVALENT):

If you have on-prem licenses with Software Assurance:

Windows Server:

- |— On-prem license → FREE Windows VM in Azure
- |— Savings: ~40% vs pay-as-you-go Windows VM
- |— Example: D4s v5 Windows VM: \$280/month → \$166/month (save \$114/month)

SQL Server:

- |— Enterprise license → Azure SQL Managed Instance at Linux price
- |— Savings: ~55% on Azure SQL Database / Managed Instance
- |— Can also apply to SQL Server on Azure VMs
- |— Example: 8-core SQL MI: \$3,200/month → \$1,440/month (save \$1,760/month!)

Linux:

- |— Red Hat / SUSE subscriptions → RHEL/SUSE VMs at lower price
- |— Savings: ~30–40%

ARCHITECT'S ADVICE:

"If your customer has existing Microsoft EA with SQL Server and Windows Server licenses, Azure is automatically 40–55% cheaper than AWS for those workloads – before any architecture optimization. This is often the single biggest factor in cloud choice for Microsoft-heavy enterprises."

Part 11 — Azure Interview Q&A: 50 Questions

Category 1: Azure Architecture & Design (15 Questions)

Q1: What is the Azure resource hierarchy and how does it differ from AWS?

A: Azure has: Entra ID Tenant → Management Groups (nested, unlimited depth) → Subscriptions → Resource Groups → Resources. Key differences from AWS: (1) Azure "Subscription" = AWS "Account" (blast radius boundary). (2) Azure has "Resource Groups" within subscriptions (AWS has no sub-account grouping). (3) Azure Management Groups support unlimited nesting (AWS OUs are more limited). (4) Azure RBAC and Policy inherit down the hierarchy (Management Group → Subscription → Resource Group → Resource). (5) Azure identity is Entra ID — shared with Office 365, Windows, on-prem AD. AWS IAM is standalone per account.

Q2: When would you choose Azure SQL Database vs SQL Managed Instance vs SQL Server on VM?

A: Decision hierarchy: (1) **Azure SQL Database** (PaaS): Best for new applications, serverless/Hyperscale options, auto-tuning, lowest operational overhead. Some SQL Server features not available (SQL Agent, cross-database queries). (2) **SQL Managed Instance** (PaaS with near-100% compatibility): Best for lift-and-shift of existing SQL Server workloads. Supports SQL Agent, linked servers, CLR, Service Broker. Best migration target when app relies on SQL Server-specific features. (3) **SQL Server on Azure VM** (IaaS): Full control, 100% compatibility, required when you need OS-level access or unsupported features. Highest operational overhead. **AWS comparison:** AWS RDS SQL Server ≈ Azure SQL Database (but with fewer features). No AWS equivalent to SQL Managed Instance. SQL Server on EC2 ≈ SQL Server on Azure VM, but Azure Hybrid Benefit makes Azure VMs 40-55% cheaper.

Q3: Explain Cosmos DB's consistency levels and when to use each.

A: Cosmos DB offers 5 consistency levels (from strongest to weakest): (1) **Strong:** All reads see the latest write globally. Use for financial transactions where

consistency is critical. Cost: highest latency. (2) **Bounded Staleness:** Reads may lag by at most K versions or T time. Use when you need predictable consistency guarantees. (3) **Session:** Consistency within a user session (most common). A user always sees their own writes. Default and recommended for most apps. (4) **Consistent Prefix:** Reads never see out-of-order writes. Use for scenarios where order matters but staleness is OK. (5) **Eventual:** No ordering guarantees, lowest latency. Use for view counters, likes, non-critical aggregations. **AWS comparison:** DynamoDB offers only Strong or Eventual — no middle ground. Cosmos DB's 5 levels let you fine-tune the consistency-performance trade-off per collection/query.

Q4: How do you design a multi-region application on Azure?

A: (1) **Azure Front Door** as the global entry point (single anycast IP, CDN, WAF, health probes). (2) **App Service or AKS** in each region behind the Front Door backend pool. (3) **Cosmos DB** with multi-region writes for global database consistency. (4) **Azure SQL** with active geo-replication or auto-failover groups for relational data. (5) **Azure Cache for Redis** per region for caching. (6) **Traffic Manager** (DNS-based) or **Front Door** (proxy-based) for failover. (7) **Event Grid / Service Bus** for cross-region event distribution. **Key Azure advantage:** Front Door provides both CDN + global LB + WAF in one service (AWS needs CloudFront + ALB + WAF + Route53). Cosmos DB active-active multi-region writes with 5 consistency levels has no AWS equivalent.

Q5: What is Azure Arc and when would you use it?

A: Azure Arc extends Azure management to any infrastructure: (1) **Arc-enabled servers:** Manage on-prem/other-cloud VMs from Azure (Azure Policy, Monitoring, Defender). (2) **Arc-enabled Kubernetes:** Manage any K8s cluster (EKS, GKE, on-prem) from Azure portal. Apply GitOps, Azure Policy, Defender. (3) **Arc-enabled data services:** Run Azure SQL Managed Instance and PostgreSQL on any K8s cluster. (4) **Arc-enabled app services:** Run Azure App Service, Functions, Logic Apps on any K8s cluster. **When to use:** Hybrid environments, multi-cloud governance, edge deployments, organizations that want Azure's management plane for non-Azure resources. **AWS comparison:** AWS Outposts runs AWS on-prem but only supports AWS services. Azure Arc manages any infrastructure and brings Azure services to any Kubernetes.

Q6: How does Azure Policy differ from AWS SCPs and Config Rules?

A: Azure Policy is more powerful because it combines what AWS needs two separate services for: (1) **Deny** effect $\approx$ SCP deny (preventive). (2) **Audit** effect $\approx$ Config Rule (detective). (3) **Modify** effect = no AWS equivalent (auto-remediate non-compliant resources — e.g., auto-add tags). (4) **DeployIfNotExists** = no AWS equivalent (auto-deploy resources — e.g., auto-configure diagnostic settings for every new resource). (5) **Append** = no AWS equivalent (modify request payload). Azure Policy can be assigned at Management Group, Subscription, or Resource Group level (like SCPs at OU/Account). It also has built-in **Policy Initiatives** (groups of policies for compliance standards: CIS, PCI, HIPAA).

Q7: What is the difference between Azure Firewall and NSGs?

A: NSGs (Network Security Groups): Layer 3/4 stateful firewall at the subnet or NIC level. Allow/deny based on IP, port, protocol. Free. $\approx$ AWS Security Groups.

Azure Firewall: Layer 3-7 managed firewall at the VNet/hub level. Supports FQDN filtering, TLS inspection (Premium), IDPS, threat intelligence, centralized logging. Costs $\sim$ \$900/month + data processing. $\approx$ AWS Network Firewall (but more mature).

When to use which: NSGs for micro-segmentation between subnets/VMs. Azure Firewall for centralized egress filtering, hub-spoke inspection, and advanced threat detection. Most enterprises use both: NSGs within spokes, Azure Firewall in the hub.

Q8-Q15: Additional Architecture Questions

#	Question	Key Points
8	How do you implement zero-trust on Azure?	Entra ID Conditional Access + PIM + Private Link + Azure Firewall + Defender
9	What is Azure Virtual WAN?	Managed hub-spoke at scale (>50 spokes). Includes VPN, ExpressRoute, Firewall. $\approx$ Managed Transit Gateway
10	How do you handle DNS in a hybrid environment?	Azure Private DNS Zones + DNS Private Resolver + conditional forwarding to on-prem
11	When to use Azure Container Apps vs AKS?	Container Apps: simple event-driven microservices (Dapr + KEDA built-in). AKS: complex K8s workloads
12	What are Availability Zones on Azure?	3+ AZs per region, zone-redundant services. Same as AWS AZs but subnets span all zones by default

#	Question	Key Points
13	How does Azure handle encryption?	SSE (default for all storage), CMK (Key Vault), Azure Confidential Computing (AMD SEV-SNP)
14	What is Azure Lighthouse?	Cross-tenant management for MSPs (manage multiple customer tenants). No AWS equivalent
15	What is Microsoft Purview?	Unified data governance: data map, data catalog, classification, lineage. ≈ Glue Data Catalog + Macie

Category 2: Migration & Modernization (10 Questions)

Q16: Walk through migrating from AWS to Azure.

A: Phase 1: **Assess** — Deploy Azure Migrate appliance in AWS (agent-based). Discover EC2, RDS, S3 workloads. Map AWS services → Azure. Phase 2: **Plan** — Deploy Azure Landing Zone, establish connectivity (ExpressRoute or site-to-site VPN between AWS VPC and Azure VNet). Phase 3: **Migrate** — VMs: Azure Migrate agent replicates EC2 → Azure VMs. Databases: DMS for SQL/PostgreSQL/MySQL. Data: AzCopy or Data Box for large S3 → Blob migrations. Phase 4: **Modernize** — EKS → AKS, Lambda → Azure Functions, DynamoDB → Cosmos DB, S3 → Blob + ADLS Gen2. **Key savings opportunity:** Apply Azure Hybrid Benefit for all Windows/SQL Server workloads immediately.

Q17: How would you migrate an on-prem .NET application to Azure?

A: Decision tree: (1) **App Service** (if web app/API — easiest path). Use App Service Migration Assistant for assessment. Deploy with deployment slots for blue/green. (2) **Azure Container Apps** (if you want containers without K8s). Containerize with Docker, deploy to Container Apps with Dapr for service communication. (3) **AKS** (if complex microservices architecture). Use Azure Migrate App Containerization tool to auto-containerize .NET apps. (4) **Azure Functions** (if event-driven/background processing). Refactor background workers to Functions. **For .NET Framework (legacy):** App Service supports .NET Framework on Windows. SQL Managed Instance supports legacy SQL features. This combination allows lift-and-shift of legacy .NET apps without code changes.

Q18-Q25: Additional Migration & Operations Questions

#	Question	Key Points
18	How to migrate Oracle to Azure?	Oracle → Azure SQL MI (with conversion) or Azure DB for PostgreSQL. Azure Hybrid Benefit doesn't apply to Oracle
19	How to set up DR on Azure?	Azure Site Recovery (VM replication), Azure SQL auto-failover groups, Cosmos DB multi-region, Traffic Manager/Front Door failover
20	How to manage secrets on Azure?	Azure Key Vault (combines secrets + keys + certificates). AKS: CSI driver for Key Vault. App Service: Key Vault references
21	How to implement GitOps on AKS?	Flux v2 (managed GitOps add-on for AKS). ≈ ArgoCD but Azure-managed
22	How to handle Windows containers?	AKS has first-class Windows node pools. AWS EKS Windows support is more limited
23	What is Azure Spring Apps?	Managed VMware Tanzu — deploy Spring Boot apps without K8s ops. No AWS equivalent
24	How to modernize a monolith on Azure?	Strangler fig → extract to App Service/Functions/Container Apps. Use Service Bus for async messaging
25	How to handle large data migrations?	Azure Data Box (100TB/1PB). AzCopy for online. Data Factory for orchestrated transfers

Category 3: Scenario-Based & Comparison (Q26-Q50)

Q26: A large bank uses Windows Server, SQL Server, and Active Directory. AWS or Azure?

A: Azure wins decisively: (1) Azure Hybrid Benefit saves 40-55% on Windows Server and SQL Server VMs (AWS charges full license cost). (2) Entra ID + AD Connect provides seamless SSO with existing Active Directory — same identity for Azure, O365, and on-prem. AWS has no Active Directory integration at this level. (3) SQL Managed Instance offers 100% SQL Server compatibility in PaaS (no AWS

equivalent). (4) Microsoft Defender + Sentinel provides enterprise SIEM/SOC (AWS needs third-party SIEM). (5) Azure DevOps provides complete ALM platform for .NET teams. For a bank with 500 Windows/SQL Server VMs, Azure Hybrid Benefit alone could save **\$500K+ per year** vs AWS.

Q27: A startup needs to build an AI chatbot. AWS Bedrock or Azure OpenAI?

A: Depends on the model: (1) **Azure OpenAI if:** You specifically want GPT-4/GPT-4o (the market leader), need enterprise content filtering, need compliance certifications for LLM usage, or the organization already uses Azure. (2) **AWS Bedrock if:** You want model diversity (Claude, Llama, Mistral), need flexibility to switch models, prefer pay-per-token without provisioning, or already on AWS. **Architect's recommendation for a startup:** Start with Azure OpenAI (GPT-4o) for the strongest model capability. Add Azure AI Search for RAG. Use Azure API Management to expose the API. Total solution deployable in 1-2 weeks. If cost becomes a concern, switch to smaller/cheaper models via Azure OpenAI's model catalog.

Q28-Q50: Rapid-Fire Q&A

#	Question	Azure Answer	AWS Comparison
28	Best service for .NET web apps?	App Service (deployment slots, auto-heal, auth built-in)	Elastic Beanstalk (inferior)
29	How to do blue-green deployments?	App Service deployment slots / AKS + Flux / Container Apps traffic splitting	CodeDeploy / ArgoCD
30	How to handle DDoS?	Azure DDoS Protection Standard (auto on all public IPs)	Shield Standard (limited) + Shield Advanced (\$3K/month)
31	How to manage multi-cloud?	Azure Arc (manage AWS, GCP, on-prem from Azure)	No equivalent
32	How to do IaC on Azure?	Bicep (Azure-native, simpler than ARM) or Terraform	CloudFormation or Terraform

#	Question	Azure Answer	AWS Comparison
33	How to queue messages?	Service Bus (enterprise messaging with sessions, FIFO, DLQ, transactions)	SQS (simpler)
34	How to handle event-driven?	Event Grid (events) + Functions (compute) + Service Bus (messaging)	EventBridge + Lambda + SQS
35	How to run Apache Spark?	Synapse Spark Pools or Azure Databricks	EMR
36	How to handle file storage (SMB)?	Azure Files (SMB 3.0) — native Windows file shares in the cloud	FSx for Windows (similar)
37	How to do API management?	Azure API Management (APIM) — policies, rate limiting, developer portal	API Gateway (less features)
38	Best SIEM for Azure?	Microsoft Sentinel — cloud SIEM + SOAR, KQL queries, 200+ connectors	No native SIEM
39	How to handle batch jobs?	Azure Batch (HPC) or Azure Functions (event-driven batch)	AWS Batch
40	How to handle IoT?	Azure IoT Hub + IoT Edge + Digital Twins	IoT Core + Greengrass
41	How to handle DevSecOps?	Defender for DevOps + GitHub Advanced Security + Azure Pipelines	CodeGuru + third-party
42	How to manage ML experiments?	Azure ML Studio (experiments, pipelines, model registry)	SageMaker Studio
43	How to handle graph databases?	Cosmos DB Gremlin API	Neptune
44	How to handle caching?	Azure Cache for Redis (Basic/Standard/Premium/Enterprise)	ElastiCache Redis
45	How to handle video streaming?	Azure Media Services + Azure CDN	MediaLive + CloudFront

#	Question	Azure Answer	AWS Comparison
46	How to handle static websites?	Azure Static Web Apps (integrated CI/CD, Functions backend)	Amplify Hosting
47	How to handle VDI/desktops?	Azure Virtual Desktop (multi-session Windows, FSLogix profiles)	WorkSpaces (single-session)
48	How to handle SAP?	SAP on Azure (certified, co-engineered, dedicated hardware)	SAP on AWS (certified)
49	How to handle compliance reporting?	Defender for Cloud regulatory compliance dashboard	Security Hub compliance
50	How to optimize reserved capacity?	Azure Advisor RI recommendations + Reservation utilization reports	Compute Optimizer + RI reports

Quick Reference – Azure Certification Path

Certification	Level	Focus
AZ-900	Foundational	Azure Fundamentals
AZ-104	Associate	Azure Administrator
AZ-204	Associate	Azure Developer
AZ-400	Expert	Azure DevOps Engineer
AZ-305	Expert	Azure Solutions Architect
AZ-500	Associate	Azure Security Engineer
AZ-700	Associate	Azure Network Engineer
DP-203	Associate	Azure Data Engineer
AI-102	Associate	Azure AI Engineer
AZ-140	Associate	Azure Virtual Desktop

Certification	Level	Focus
SC-100	Expert	Cybersecurity Architect
SC-200	Associate	Security Operations Analyst (Sentinel)

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